PEMEtrexed and CARBOplatin Therapy

Please refer to NCCP Regimen 00849 Nivolumab 360mg and Chemotherapy for relevant information when used in combination with nivolumab

INDICATIONS FOR USE:

INDICATION	ICD10	Regimen Code	HSE approved reimbursement status*
Treatment of chemotherapy naïve patients with unresectable malignant pleural mesothelioma[i].	C45	00318a	N/A
First line treatment of patients with locally advanced or metastatic non-small cell lung cancer (NSCLC) [ii].	C34	00318b	N/A
In combination with nivolumab for the neoadjuvant treatment of resectable NSCLC at high risk of recurrence in adult patients whose tumours have PD-L1 expression ≥ 1% **(3 cycles only)** **(This combination is available in NCIS (00849.4))**	C34	00318c	Nivolumab: ODMS 01/05/2024 Chemotherapy: N/A

*This applies to post 2012 indications

TREATMENT:

The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patients individual clinical circumstances.

For the treatment of malignant pleural mesothelioma PEMEtrexed and CARBOplatin are administered once every 21 days for a maximum of 4-6 cycles unless the patient experiences disease progression or unacceptable toxicity.

For the first line treatment of locally advanced or metastatic NSCLC PEMEtrexed and CARBOplatin are administered once every 21 days for 4-6 cycles followed by maintenance PEMEtrexed monotherapy (Reference NCCP regimen 00222) unless the patient experiences disease progression or unacceptable toxicity.

Facilities to treat anaphylaxis MUST be present when systemic anti-cancer therapy (SACT) is administered.

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Admin. Order	Day	Drug	Dose	Route	Diluent & Rate	Cycle
1	1	PEMEtrexed	500mg/m^2	IV infusion	100mL NaCl 0.9% over 10 minutes	Every 21 days
2	1	CARBOplatin	AUC 5	IV infusion	500mL Glucose 5% over 30 minutes	Every 21 days
		Folic Acid or multivitamin containing 350-1000 micrograms folic acid	350-1000 micrograms[a]	PO		
[a] At least five doses of folic acid must be taken during the seven days preceding the first dose of PEMEtrexed, and dosing must continue during the full course of therapy and for 21 days after the last dose of PEMEtrexed. See Premedications for further treatment required.						
PEMEtrexed is physically incompatible with diluents containing calcium, including lactated Ringer's injection and Ringer's injection.						

Note: Administration volumes and fluids have been standardised to facilitate electronic prescribing system builds.

CARBOplatin dose:

The dose in mg of CARBOplatin to be administered is calculated as follows:

Dose (mg) = target AUC (mg/mL x minute) x (GFR mL/minute +25)

- **Measured GFR** (e.g. nuclear renogram) is preferred whenever feasible
- Estimation of GFR may be performed using the Wright formula to estimate GFR or the Cockcroft and Gault formula to estimate creatinine clearance
- The GFR used to calculate the AUC dosing should not exceed 125mL/minute
- For obese patients and those with a low serum creatinine, for example, due to low body weight or post-operative asthenia, estimation using formulae may not give accurate results; measured GFR is recommended
    - where obesity (body mass index [BMI] ≥ 30 kg/m^2) or overweight (BMI 25-29.9) is likely to lead to an overestimate of GFR and isotope GFR is not available the use of the adjusted ideal body weight in the Cockcroft and Gault formula may be considered
    - where serum creatinine is less than 63 micromol/L, the use of a creatinine value of 62 micromol/L or a steady pre-operative creatinine value may be considered
- These comments do not substitute for the clinical judgement of a physician experienced in prescription of CARBOplatin

WRIGHT FORMULA

There are two versions of the formula depending on how serum creatinine values are obtained, by the kinetic Jaffe method or the enzymatic method. The formula can be further adapted if covariant creatine kinase (CK) values are available (not shown).

1. *SCr measured using enzymatic assay.*

$$\text{GFR (mL/min)} = \frac{(6230 - 32.8 \times \text{Age}) \times \text{BSA} \times (1 - 0.23 \times \text{Sex})}{\text{SCr (micromol/minute)}}$$

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2. *SCr measured using Jaffe assay*

$$\text{GFR (mL/min)} = \frac{(6580 - 38.8 \times \text{Age}) \times \text{BSA} \times (1 - 0.168 \times \text{Sex})}{\text{SCr (micromol/minute)}}$$

Key: Sex = 1 if female, 0 if male; Age in years; BSA= DuBois BSA

COCKCROFT-GAULT FORMULA

$$\text{GFR (mL/min)} = \frac{S \times (140 - \text{age in years}) \times \text{wt (kg)}}{\text{Serum creatinine (micromol/L)}}$$

S= 1.04 for females and 1.23 for males

ELIGIBILITY:

- Indications as above
- ECOG 0-2

EXCLUSIONS:

- Hypersensitivity to PEMEtrexed, CARBOplatin* or any of the excipients
- PEMEtrexed
  - Creatinine clearance < 45mL/minute
- CARBOplatin
  - Pre-existing neuropathies ≥ grade 2
  - Significant hearing impairment/tinnitus
- Pregnancy
- Breastfeeding

*If it is felt that the patient may have a major clinical benefit from CARBOplatin, it may in exceptional circumstances be feasible to rechallenge a patient with a prior mild hypersensitivity reaction e.g. using a desensitisation protocol, but only with immunology advice, premedication as advised, and a desensitisation protocol under carefully controlled conditions with resuscitation facilities available and medical and/or ITU/ HDU supervision

PRESCRIPTIVE AUTHORITY:

The treatment plan must be initiated by a Consultant Medical Oncologist.

TESTS:

Baseline tests:

- FBC, renal and liver profile
- Isotope GFR measurement (preferred) or GFR / Clearance estimation
- Audiology referral as clinically indicated

Regular tests:

- FBC, renal and liver profile prior to each cycle

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Disease monitoring:
Disease monitoring should be in line with the patient's treatment plan and any other test/s as directed by the supervising Consultant.

DOSE MODIFICATIONS:

- Any dose modification should be discussed with a Consultant

Haematological:

- Treatment may be delayed to allow sufficient time for recovery
- Upon recovery patient should be retreated using the guidelines below

Table 1: Dose modifications for haematological toxicity of PEMEtrexed and CARBOplatin

ANC x 10^9 /L	Comment	Platelets x 10^9 /L	Comment
≥ 1.5	100%	≥100	Continue
1.0 to < 1.5	Consult with team, if clinically well continue treatment	75 - <100	Consult with team, if clinically well, consider continuation
0.5 to < 1.0	Delay until treatment recovery	50 - < 75	Delay until treatment recovery
< 0.5	Delay until treatment recovery and reduce PEMEtrexed and CARBOplatin by 25%	<50	Delay until treatment recovery and reduce PEMEtrexed and CARBOplatin by 25%
Febrile neutropenia	Delay until treatment recovery and reduce PEMEtrexed and CARBOplatin by 25%	<50 with bleeding	Delay until treatment recovery and reduce PEMEtrexed and CARBOplatin by 50%

Renal and Hepatic Impairment:

Table 2: Dose modification of CARBOplatin and PEMEtrexed in renal and hepatic impairment

	Renal Impairment		Hepatic Impairment	
CARBOplatin[a]	See note below*		No dose modification required	
PEMEtrexed[b]	**CrCl (mL/min)**	**Dose**		**Dose**
	≥45	No dose adjustment is needed	Mild and moderate	No need for dose adjustment is expected
	<45	Not recommended	Severe	Not recommended, based on the risk of PEMEtrexed induced liver dysfunction.
	Haemodialysis	Not recommended		

[a] CARBOplatin (renal and hepatic –NCCP standardisation)
[b] PEMEtrexed renal and hepatic as per Giraud et al 2023

***Renal dysfunction and CARBOplatin:**

- Patients with creatinine clearance (CrCl) values of <60mL/minute are at greater risk of developing myelosuppression
- If GFR between 20 to ≤ 30mL/minute, CARBOplatin should be administered with extreme caution
- If GFR ≤ 20mL/minute CARBOplatin should not be administered at all
- If Cockcroft & Gault or Wright formula are used, the dose should be calculated as required per cycle based on a serum creatinine obtained within 48 hours of drug administration

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If isotope GFR is used, the dose can remain the same provided the serum creatinine is ≤110% of its value at the time of the isotope measurement. If the serum creatinine increases, consideration should be given to remeasuring the GFR or to estimating it using Cockcroft & Gault or Wright formulae.

Management of adverse events:

Table 3: Dose Modification of PEMEtrexed and CARBOplatin for Adverse Events

Adverse effect	Recommended dose modification
Diarrhoea Grade ≥ 3 and any diarrhoea irrespective of grade requiring hospitalisation	Withhold treatment until resolution to less than or equal to patient's pre-therapy value. Resume at 75% of previous dose for both PEMEtrexed and CARBOplatin.
Mucositis Grade ≥3	Withhold treatment until resolution to less than or equal to patient's pre-therapy value. Resume at 50% of previous dose for PEMEtrexed and at 100% of previous dose for CARBOplatin.
Neurotoxicity Grade 2 Grade ≥3 or any recurrence of neurotoxicity	Withhold treatment until resolution to less than or equal to patient's pre-therapy value. Resume at 100% dose for PEMEtrexed and reduce CARBOplatin by 50%. Discontinue
Other Grade ≥3	Withhold treatment until resolution to less than or equal to patient's pre-therapy value. Resume at resume at 75% of previous dose PEMEtrexed and CARBOplatin.
Grade ≥3 toxicity after 2 dose reductions	Consider discontinuation of both drugs. Discuss with consultant.
Allergic reactions: Grade 3 ≥	Clinician decision. Consider discontinuation of whichever drug caused the allergy.

SUPPORTIVE CARE:

EMETOGENIC POTENTIAL

- As outlined in NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting - Available on the NCCP website

PEMEtrexed: Low (**Refer to local policy)**
CARBOplatin: High (**Refer to local policy)**

For information:
Within NCIS regimens, antiemetics have been standardised by the Medical Oncologists and Haemato-oncologists. Information is available in the following documents:

- NCCP Supportive Care Antiemetic Medicines for Inclusion in NCIS (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for Inclusion in NCIS (Haemato-oncology) - Available on the NCCP website

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PREMEDICATIONS:

- A corticosteroid should be given the day prior to, on the day of, and the day after PEMEtrexed administration. The corticosteroid should be equivalent to 4 mg of dexAMETHasone administered orally twice a day
- Intramuscular injection of vitamin B_{12} (hydroxycobolamin) (1000 micrograms) in the week preceding the first dose of PEMEtrexed and once every three cycles thereafter. Subsequent vitamin B_{12} injections may be given on the same day as PEMEtrexed

OTHER SUPPORTIVE CARE: No specific recommendations

ADVERSE EFFECTS

- Please refer to the relevant Summary of Product Characteristics (SmPC) for details

DRUG INTERACTIONS:

- Current SmPC and drug interaction databases should be consulted for information.

REFERENCES:

1. Ceresoli GL. Zucali PA et al. Phase II study of pemtrexed plus carboplatin in Malignantpleual mesothelioma. J ClinOncol 2006;24 (9);1443-1447
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3. Forde PM et al; CheckMate 816 Investigators. Neoadjuvant Nivolumab plus Chemotherapy in Resectable Lung Cancer. N Engl J Med. 2022 May 26;386(21):1973-1985. doi: 10.1056/NEJMoa2202170. Epub 2022 Apr 11. PMID: 35403841; PMCID: PMC9844511
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6. Giraud E L, Lijster B D, et al. Dose recommendations for anticancer drugs in patients with renal or hepatic impairment: an update. Available at: https://www.thelancet.com/journals/lanonc/article/PIIS1470-2045(23)00216-4/fulltext
7. NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting. V6 2025. Available at: https://www.hse.ie/eng/services/list/5/cancer/profinfo/chemoprotocols/nccp-classification-document-for-systemic-anti-cancer-therapy-sact-induced-nausea-and-vomiting.pdf
8. PEMEtrexed (ALIMTA®) Summary of Product Characteristics. Last updated: 05/03/2025. Accessed July 2025. Available at: https://www.ema.europa.eu/en/documents/product-information/alimta-epar-product-information_en.pdf
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Version	Date	Amendment	Approved By
1			Prof Maccon Keane
2	02/05/2018	Applied new NCCP regimen template, updated dosing in hepatic impairment, adverse reactions and drug interactions	Prof Maccon Keane
3	29/04/2020	Reviewed. Update of exclusions and adverse events.	Prof Maccon Keane

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4	29/09/2023	Updated CARBOplatin infusion time. Updated standard wording for CARBOplatin dosing, renal dysfunction and creatinine value. Updated baseline tests and exclusions section.	Prof Maccon Keane
5	01/05/2024	New indication for nivolumab in the neoadjuvant setting and reference to relevant nivolumab regimen added.	Prof Maccon Keane
6	17/10/2025	Regimen reviewed. Updated: - Treatment outline - Exclusion criteria - Baseline testing - Haematological dose modifications – removed nadir blood counts - Renal and hepatic dose modifications to align with Giraud et al 2023 - Dose modification recommendations for non-haematological toxicities. - Dose modifications in adverse events - Updated regimen in line with NCCP standardisation	Prof Maccon Keane

Comments and feedback welcome at oncologydrugs@cancercontrol.ie.

[i] This is an unlicensed indication for the use of PEMEtrexed and CARBOplatin in Ireland. Patients should be informed of this and consented to treatment in line with the hospital's policy on the use of unlicensed medication and unlicensed or "off label" indications. Prescribers should be fully aware of their responsibility in communicating any relevant information to the patient and also ensuring that the unlicensed or "off label" indication has been acknowledged by the hospital's Drugs and Therapeutics Committee, or equivalent, in line with hospital policy.

[ii] This is an unlicensed indication for the use of PEMEtrexed and CARBOplatin in Ireland. Patients should be informed of this and consented to treatment in line with the hospital's policy on the use of unlicensed medication and unlicensed or "off label" indications. Prescribers should be fully aware of their responsibility in communicating any relevant information to the patient and also ensuring that the unlicensed or "off label" indication has been acknowledged by the hospital's Drugs and Therapeutics Committee, or equivalent, in line with hospital policy.

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